

Trainer Debrief — Lab 6

SIPOC, Process Mapping and Scope Governance

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Map your capstone process and govern its scope through the charter (A2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-06-sipoc-process-map.xlsx

Total cycle time is 44 min against 227 min of wait time — process cycle efficiency = $44/271 = 16\%$, so 84% of lead time is pure waiting. Only 'Run weld cycle' and 'Rework and re-test' transform the product; the burst test is BVA (required by the regulator, adds no customer value); every transfer and queue is NVA. There are five HANDOFFS, and the two biggest wait times sit immediately after them. The scope sheet is what stops scope creep in week three: without a written start point ('fixture load') and end point ('final disposition'), the team will be asked to fix incoming material quality too.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your SIPOC's first and last steps match your charter scope exactly, your swimlane map marks every handoff and rework loop, and the map has been validated by walking the actual process.

Where this sits in the capstone

Re-derive SIPOC and detailed process mapping in full, then apply them to bound your capstone. Scope creep kills more Black Belt projects than bad statistics — this lab builds the map that makes scope arguments objective. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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